



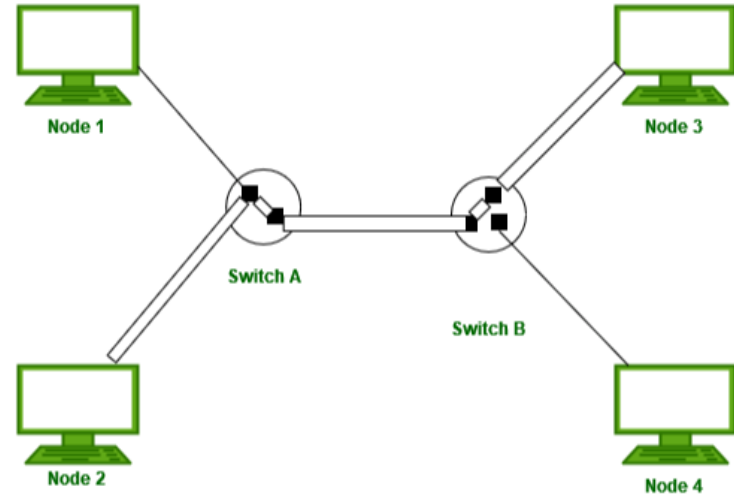
Computer Networks

PPT-8 Network layer

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Switching

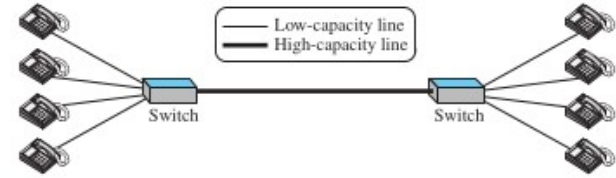
Network in which a switch connects at least two links together. A switch needs to forward data from a network to another network when required.



Circuit-Switched Network

In a circuit-switched network, a dedicated connection, called a circuit, is always available between the two end systems; the switch can only make it active or inactive.

Public Switched Telephone Network (PSTN) is a classic example of a circuit-switched network.



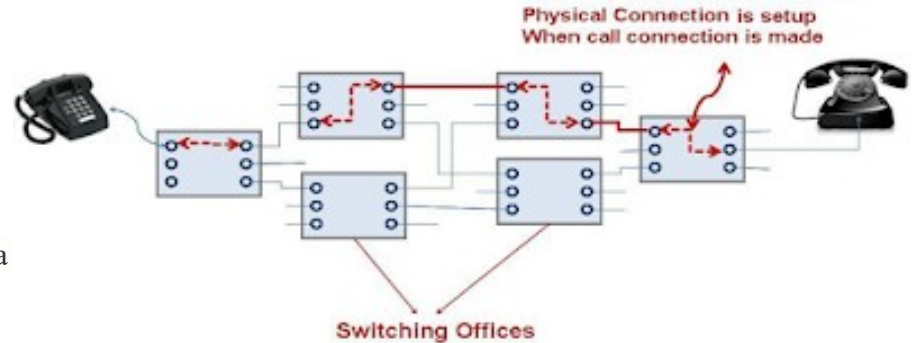
Circuit-Switched Network

Advantages:

- consistent bandwidth and low latency.
- suitable for real-time applications like voice calls.

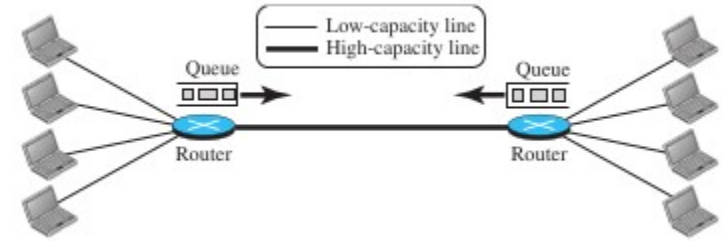
Disadvantages:

- Connection setup and termination generate extra traffic.
- User pay for circuit, even when not sending data.
- Other user cannot use circuit even if it is idle / free.

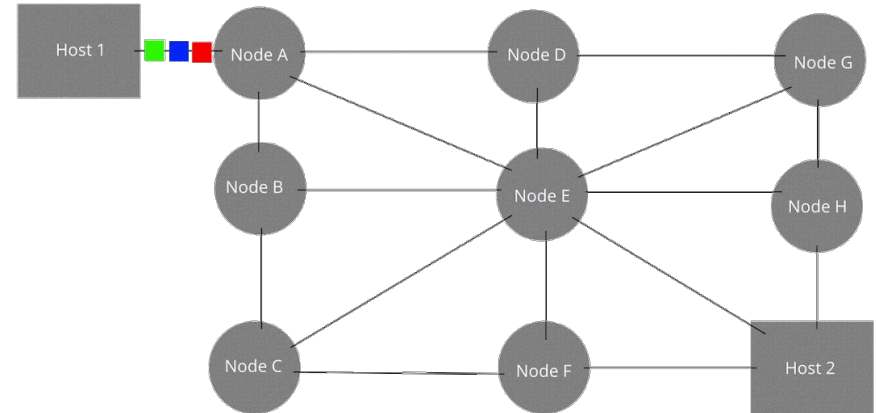


Packet-Switched Network

Data exchange between two endpoints happens in small chunks called **packets**. Instead of a continuous, uninterrupted connection like a telephone call, computers send and receive these individual packets separately. Because each packet is independent, network switches can store them temporarily and forward them when needed, enabling efficient data transmission.



The original message is Green, Blue, Red.

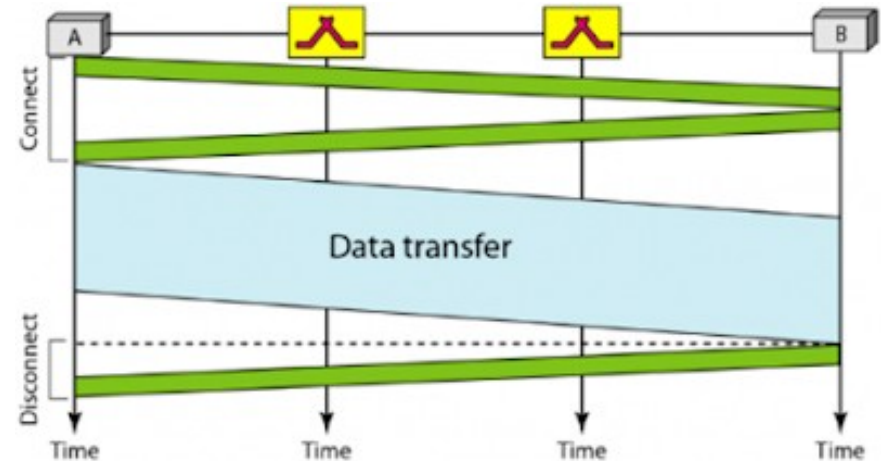
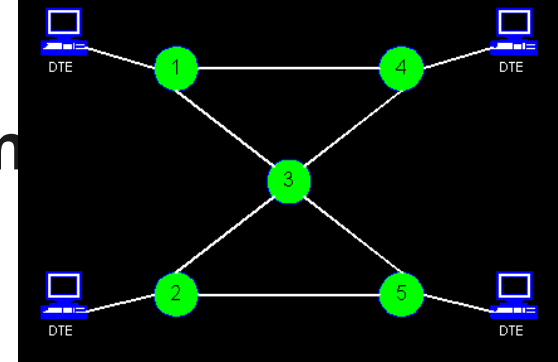


Connection oriented packet switching

1. Logical path established before data transfer
2. All packets follow the same route

Phases:

- **Setup:** Connection established, QoS decided
- **Data Transfer:** Packets sent with flow & error control
- **Release:** Connection terminated after transmission



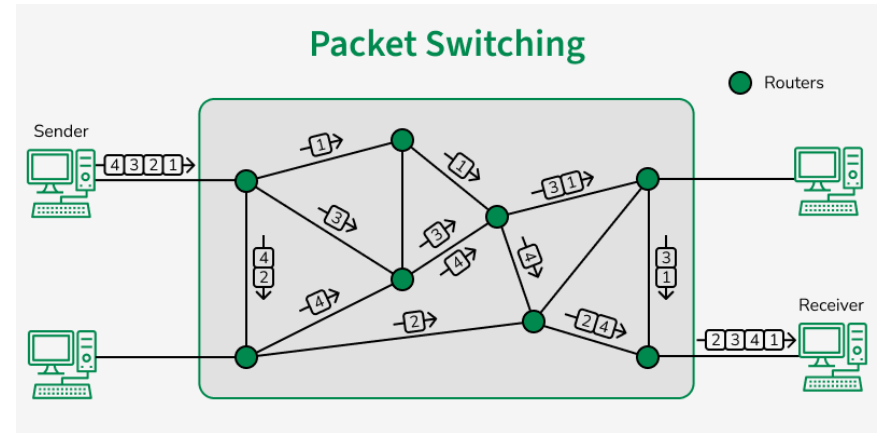
Packet-Switched Network

Advantages:

- Resource utilization more efficiently.
- It is more flexible.
- Small size packet reduces transmission delay.
- Small amount of time required to establish or terminate circuit.

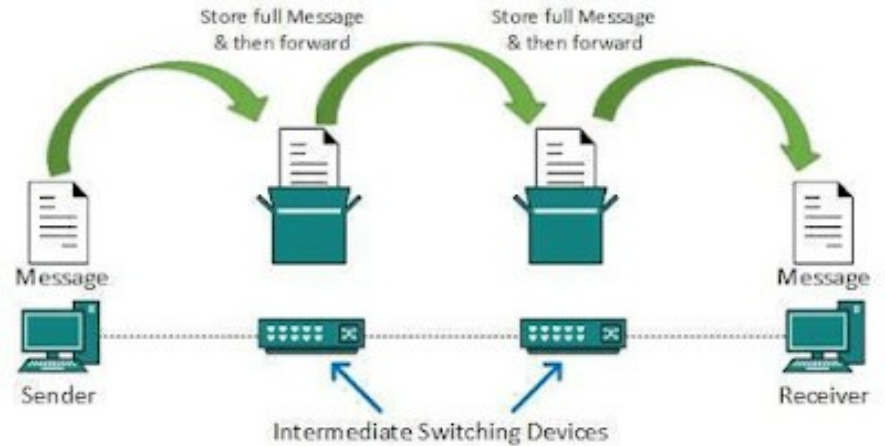
Disadvantages:

- Complex protocol for packet switching.
- Algorithms are more complicated.
- Switching processor must be powerful.
- Packet loss may occur during switching.



Message Switching

Message switching is used to describe the telegraph network. No need to establish dedicated path. When the sender has a block of data to be sent, it is stored in router and forward to next hop. Each block is received in its entirety, inspected for errors, and then transmitted. A network using this technique is called store and forward network.





Message Switching

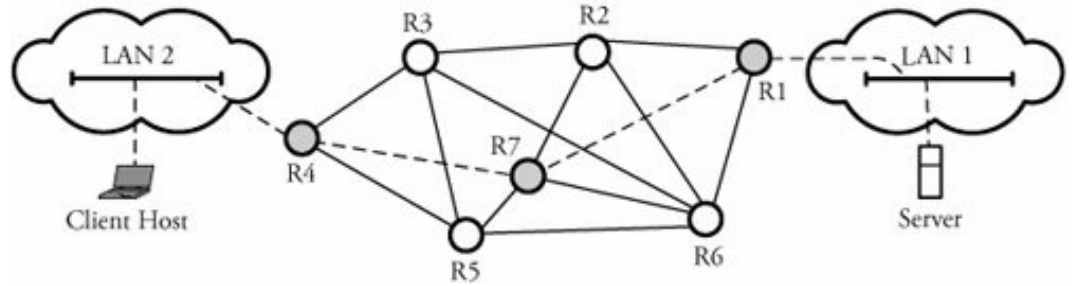
Advantages:

- Efficient traffic management.
- Reduce network traffic congestion.
- Efficient use of traffic channel.

Disadvantages:

- Store and forward technique increase delay in data transmission.
- Each node requires large capacity for storing.

Network Layer



1. The network layer is responsible for end-to-end delivery of packets between the source and destination hosts across one or more networks. It handles host-to-host communication and makes sure packets can travel through intermediate routers to reach their destination.
2. Because data may pass through many routers, the network layer's job includes selecting the best path for each packet and **routing** it accordingly.
3. Separating this routing duty from the transport layer allows routers to operate without needing transport or application protocols, simplifying network design.



Core Functions

1. **Logical Addressing:** Assigns and uses IP addresses so each device is uniquely identified on the network.
2. **Routing & Forwarding:** Uses routers and routing tables to choose and follow optimal paths for packet delivery.
3. **Packet Delivery:** Moves packets from source to destination, even across multiple networks.
4. **Connectionless Service:** Protocols like IP don't establish a connection before sending data and don't provide flow, error, or congestion control — these are handled by higher layers if needed.
5. **IP (Internet Protocol)** operates at the network layer and is responsible for delivering packets called **datagrams** from the source host to the destination host. IP itself is **connectionless** and does not guarantee delivery.



Auxiliary Support Protocols:

- **ICMP:** Reports issues like unreachable destinations or errors.
- **IGMP:** Helps manage multicast group membership.
- **ARP:** Resolves logical IP addresses to physical MAC addresses on a local network.
- **DHCP:** Provides IP addresses dynamically to hosts.



Routing Algorithm

- Dijkstra's Algorithm
- Bellman-Ford Algorithm



Difference Between Connection-oriented and Connection-less Services